AP BIOLOGY TOPIC 2.7

Tonicity, Water Potential (Ψ) & Osmoregulation Strategies

UNIT 2: CELL STRUCTURE AND FUNCTION (10–13% EXAM WEIGHT)

ENE-2.H | ENE-2.I | Ψ = ΨP + ΨS | ΨS = −iCRT

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 2.7 (TONICITY, WATER POTENTIAL (Ψ) & OSMOREGULATION STRATEGIES)

CED Core Vocabulary

Osmosis — Passive diffusion of free water (H_2O) across a selectively permeable membrane (mainly through aquaporins); requires NO ATP .

2. Solute Concentration — From LOW osmolarity (solute concentration) → HIGH osmolarity (solute concentration) ("Solute sucks!") .

3. Tonicity — From a HYPOTONIC solution → HYPERTONIC solution across a membrane.

X-intercept (where % Change in Mass = 0%) — Gives the isotonic molarity (C) of the potato cells! Plug that C into Ψs = −iCRT to calculate the potato's solute potential!

In an Open Beaker / Animal Cell — Ψp = 0 bars (at atmospheric pressure).

In a Turgid Plant Cell — Ψp > 0 (positive turgor pressure exerted by cell wall pushing back).

In Plant Xylem Vessels (Transpirational Pull) — Ψp < 0 (negative tension/suction).

Pure Water in Open Beaker — Ψs = 0 and Ψp = 0 → Ψ = 0 bars (MAXIMUM possible water potential at 1 atm!) .

§2. OSMOSIS & TONICITY: DIRECTION OF WATER MOVEMENT

Passive H_2O Flow

- **Osmosis:** Passive diffusion of **free water (** H_2O **)** across a selectively permeable membrane (mainly through **aquaporins**); requires **NO ATP**.
- **Universal Rule for Direction of Net Water Flow (3 Equivalent Ways to State It):**
  - **1. Water Potential (** Ψ **):** From **HIGH water potential (** Ψ **) → LOW (more negative) water potential (** Ψ **)**.
  - **2. Solute Concentration:** From **LOW osmolarity (solute concentration) → HIGH osmolarity (solute concentration)** *("Solute sucks!")*.
  - **3. Tonicity:** From a **HYPOTONIC solution → HYPERTONIC solution** across a membrane.

[AP EXAM FRQ SCORING STRATEGY] **FRQ Precision Tip:** Always state that water moves by osmosis **"from a region of higher water potential (** Ψ **) / lower solute concentration to a region of lower water potential (** Ψ **) / higher solute concentration"** — don't just say "toward solute"!

EXTERNAL SOLUTION TONICITY	NET H_2O DIRECTION	ANIMAL CELL (NO CELL WALL)	PLANT CELL (RIGID CELL WALL)
Hypotonic Solution *(Lower solute outside; higher* **Ψ** *outside)*	Net H_2O **ENTERS cell**	**Lysed (Bursts / Cytolysis)**; swells until plasma membrane ruptures	**Turgid (Normal / Ideal!)**; central vacuole swells & pushes plasma membrane against cell wall (**+Ψp**)
Isotonic Solution *(Equal solute & equal* **Ψ** *inside/outside)*	**No net** H_2O **movement** (equal rates in & out)	**Normal (Ideal!)**; e.g., RBCs in **0.9% NaCl** saline	**Flaccid (Limp)**; no turgor pressure (**Ψp = 0**), plant wilts
Hypertonic Solution *(Higher solute outside; lower/more neg* **Ψ** *outside)*	Net H_2O **LEAVES cell**	**Shriveled (Crenated)**; dehydrates & shrinks	**Plasmolyzed (Wilted/Lethal)**; plasma membrane pulls away from cell wall as vacuole shrinks

§4. WATER POTENTIAL (Ψ) & SOLUTE POTENTIAL EQUATIONS

AP Math Must-Know

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Water Potential (Ψ) & Solute Potential Equations ➔ **Pressure Potential (Ψp)** ➔ **Solute Potential (Ψs, Osmotic Potential)** ➔ **Variables in Ψs = −iCRT**

- **Water Potential (** Ψ **, measured in bars or MPa):** Quantifies the potential energy of water to move from one area to another due to osmosis, gravity, and mechanical pressure.

Ψ = Ψp + Ψs | Ψs = −iCRT

- **Pressure Potential (** Ψp **):** Physical/hydrostatic pressure on solution:
  - **In an Open Beaker / Animal Cell:** **Ψp = 0 bars** (at atmospheric pressure).
  - **In a Turgid Plant Cell:** **Ψp > 0** (positive turgor pressure exerted by cell wall pushing back).
  - **In Plant Xylem Vessels (Transpirational Pull):** **Ψp < 0** (negative tension/suction).
- **Solute Potential (** Ψs **, Osmotic Potential):** Effect of dissolved solutes on water potential:
  - **Pure Water in Open Beaker:** **Ψs = 0** and **Ψp = 0** → **Ψ = 0 bars** **(MAXIMUM possible water potential at 1 atm!)**.
  - **Adding Solute ALWAYS Lowers** **Ψs** **:** Solute binds water molecules via hydration shells, reducing free water → **Ψs** **is ALWAYS** **0** **or NEGATIVE (** **≤ 0** **)**. More solute = more negative **Ψs** !
- **Variables in** **Ψs = −iCRT** **:**
  - **i** **= Ionization constant** (number of particles formed per formula unit in water):
    - **Covalent non-electrolytes (Glucose, Sucrose):** Do NOT ionize → **i = 1.0** .
    - **Ionic salts:** **$NaCl \rightarrow Na^+ + Cl^-$** (**i = 2**); **$CaCl_2 \rightarrow Ca^{2+} + 2Cl^-$** (**i = 3**).
  - **C** **= Molar concentration** (**mol/L** or **M**).
  - **R** **= Pressure constant** = **0.0831 liter·bar / (mol·K)** .
  - **T** **= Absolute temperature in Kelvin** (**K = °C + 273**).

[EXAMPLE & BIOLOGICAL APPLICATION] **Worked AP Exam Calculation**

Example:

Find the water potential (**Ψ**) of an **open beaker** containing **0.3 M** **sucrose** at **27°C** :

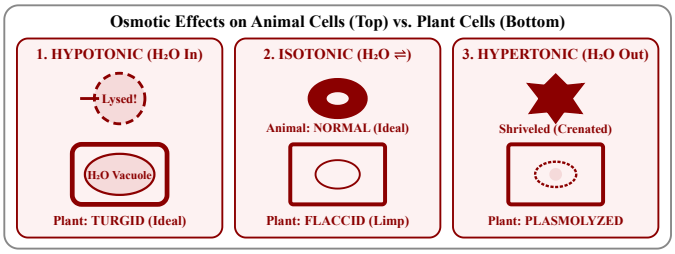
1. Convert Temp: **T = 27 + 273 = 300 K** ; for sucrose, **i = 1.0** ; open beaker **Ψp = 0** .
2. Calculate **Ψs = −(1.0)(0.3 mol/L)(0.0831 L·bar/mol·K)(300 K) = −7.48 bars** .
3. Total **Ψ = Ψp + Ψs = 0 + (−7.48) = −7.48 bars** . If a plant cell at equilibrium in this beaker has **Ψs = −9.0 bars** , its turgor pressure is **Ψp = Ψ − Ψs = −7.48 − (−9.0) = +1.52 bars** !

[CORE AXIOM / REMEMBER] **Golden Rules of Water Potential (** **Ψ** **):**

- **Pure water in an open container has** **Ψ = 0** .
- **More dissolved solute → more negative** **Ψs** **→ lower total** **Ψ** .
- Water ALWAYS moves passively toward the **more negative** **Ψ** (e.g., from **Ψ = −1.5 bars** to **Ψ = −4.0 bars**)!

§3. TONICITY IN ANIMAL VS. PLANT CELLS VISUALIZED

Osmotic States



[AP EXAM FRQ SCORING STRATEGY] **Classic AP Biology Potato Core Osmosis Lab (FRQ Favorite!):**

- Potato cores in **Hypotonic** sucrose solutions → gain water → **% Change in Mass is POSITIVE (+)**.
- Potato cores in **Hypertonic** sucrose solutions → lose water → **% Change in Mass is NEGATIVE (−)**.
- **X-intercept (where % Change in Mass = 0%):** Gives the **isotonic molarity (C)** of the potato cells! Plug that **C** into **$\Psi s = -iCRT$** to calculate the potato's solute potential!

§5. OSMOREGULATION: MAINTAINING WATER & SOLUTE BALANCE

Organism Adaptations

- **Osmoregulation:** Active regulation of internal water volume and solute concentrations (osmolarity) to maintain **homeostasis, cell volume survival, and optimal enzyme conformation**.

ORGANISM / HABITAT	OSMOTIC CHALLENGE	KEY OSMOREGULATORY ADAPTATIONS
Freshwater Protists & Fish *(Hypotonic environment)*	Water constantly **enters** cells by osmosis; salts lost by diffusion	**Contractile vacuole** (in *Paramecium*/*Amoeba*) uses ATP to pump water out; freshwater fish **do not drink**, excrete large volumes of **very dilute urine**, & gills actively uptake ions
Marine (Saltwater) Bony Fish *(Hypertonic environment)*	Water constantly **leaves** body via gills; salt diffuses in	Constantly **drink seawater**, actively pump excess **Na^+/Cl^-** out across gill chloride cells, & excrete tiny amounts of **concentrated urine**
Terrestrial Animals	Dehydration via evaporation & excretion	Waterproof skin/cuticle; **mammalian kidneys (Loop of Henle + ADH)** reabsorb water to produce hyperosmotic urine
Vascular Plants	Transpirational water loss vs. turgor support	Central vacuole accumulates solutes (lowers **Ψs**) to draw water in; guard cells **close stomata** during drought (via hormone ABA)

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 2.7 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 2.7: Tonicity, Water Potential (Ψ) & Osmoregulation Strategies ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigation 4 (Diffusion & Osmosis): Dialysis tubing & potato core mass change (% ΔMass = [(Final − Initial) / Initial] × 100) across sucrose molarities (0.0 M to 1.0 M). IV: Extracellular sucrose molarity (M); DV: % change in mass; Isotonic point: x-intercept where % ΔMass = 0% (no net water movement). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Graph Rules: Total Water Potential: $\Psi = \Psi_p + \Psi_s$. Solute Potential: $\Psi_s = -iCRT$ (where i = ionization constant [1 for sucrose/glucose, 2 for NaCl], C = molarity in mol/L, R = 0.0831 L·bar/mol·K, T = K = °C + 273). Surface-Area-to-Volume Ratio (SA:V): Cube SA = $6s^2$, V = s^3 (SA:V = 6/s). Larger SA:V accelerates diffusion per unit volume. **Error Bar Rule (±2 SE_x̄):** If the ±2 $SE_{\bar{x}}$ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Physiological Disruption Case Study: Cystic Fibrosis results from a deletion (ΔF508) in the CFTR chloride channel protein, preventing misfolded channels from reaching the apical plasma membrane. Impaired Cl^- efflux prevents osmotic water movement into mucus, producing thick, dehydrated mucus in lungs and pancreatic ducts. Ouabain/Digoxin inhibits the Na^+/K^+-ATPase pump, collapsing the electrochemical gradient required for secondary active cotransport.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 2.7 to prove why the claim holds.

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